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Building a Model Factory

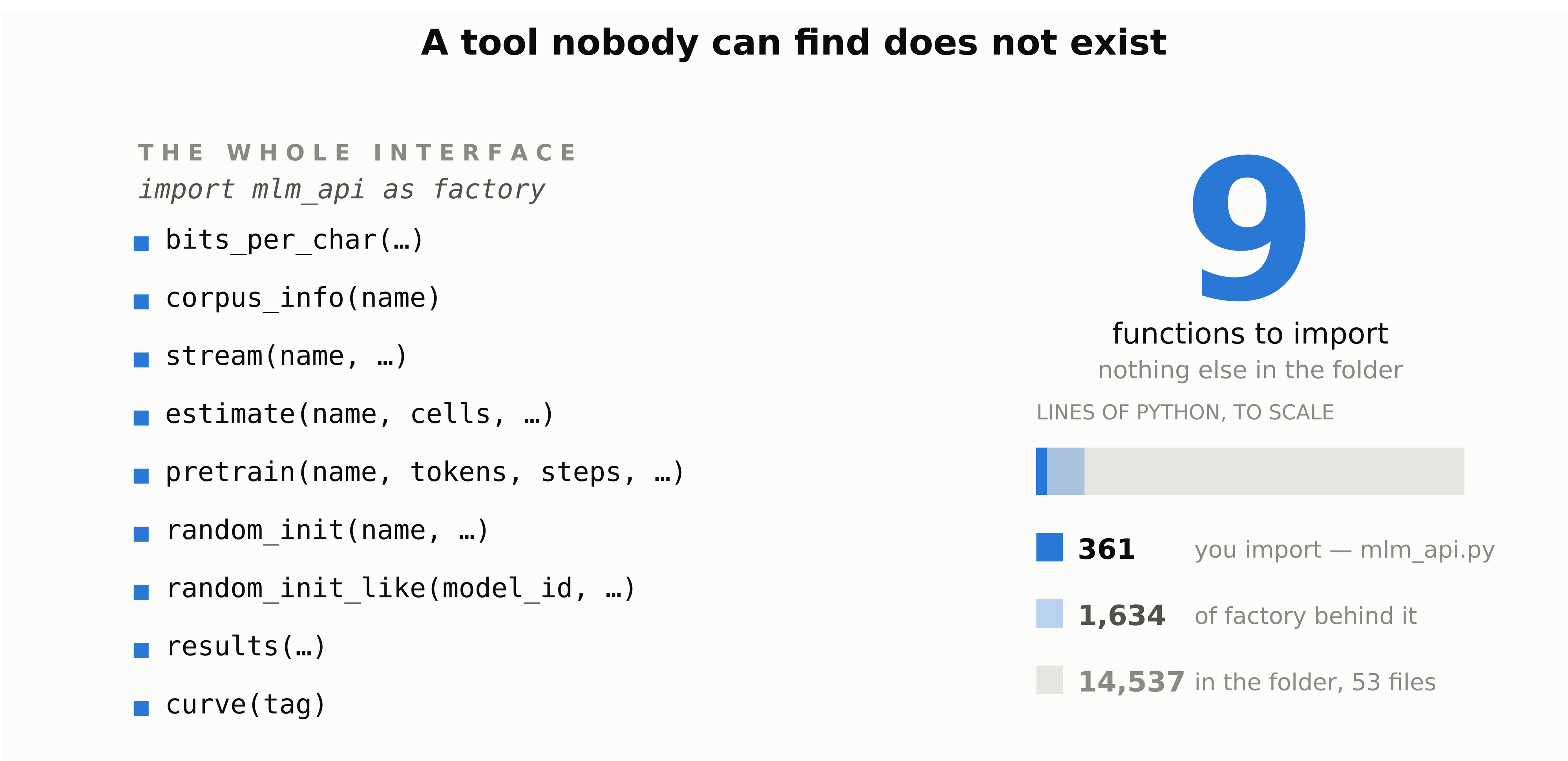
From one-off notebooks to repeatable, challengeable experiments

A2-NLP Project Team · CSED 504 · University of Washington · Summer 2026

TAKEAWAY

The factory made hundreds of model comparisons affordable and auditable—then gave collaborators enough control to disprove results that initially looked convincing.

ONE SCREEN TO OPERATE THE FACTORY



Leon cloned the repository, read the documentation, and asked whether there was an interface he was supposed to be using. There was. The folder's front page was titled with a different study and mlm_api first appeared on line 25, one row of a second table. Every word of it accurate. Still unfindable — and that is a failure of the interface, not of the reader.

9

public MLM API functions

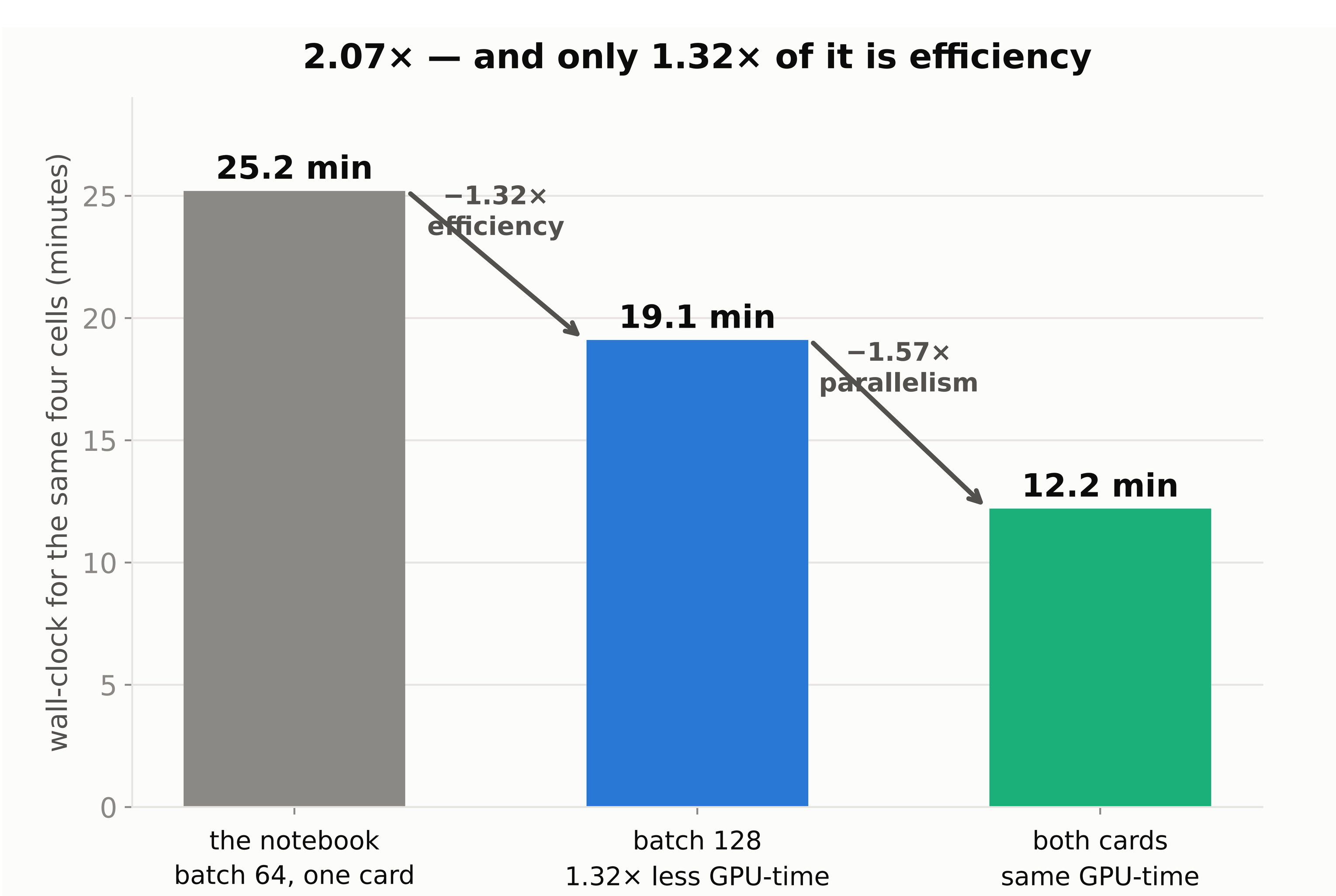
2.07×

faster end-to-end · 1.32× efficiency

197

pretraining run records

ITERATION BECAME AFFORDABLE



Both changes were available on day one and neither is clever. The batch was simply too small, and the second card had never been used. Utilization went from one card busy and one idle to 91% and 93%.

The operating rule

Prepare once → inspect and fingerprint → estimate on the actual GPU → pretrain one cell or queue a fleet → read curves and results through the same API. Interactive exploration stays in notebooks; expensive work moves to restartable processes.

On the Yoruba corpus, reading, fitting the BPE, encoding 260M characters and moving the flat token store to GPU took 53 seconds. One 98M-parameter training run took 85 minutes—a 96× ratio that defines the boundary between interactive work and unattended queues.

Identity

Every setting that can move a number belongs in the record. Corpora and vocabularies carry fingerprints; run reuse refuses mismatched settings. The current evidence base contains 197 pretraining runs, 278 fine-tuning records and 892 individual fine-tuning runs.

Statistical guardrails

At three seeds per arm, a mean difference must be about 2.27× the pooled sample spread to clear a two-sided 0.05 t-threshold. An exact permutation test cannot return below p=0.10 with 3 vs 3. A pre-registered sample size limits what a run grid is allowed to claim.

Failure is evidence

Healthy and doomed runs overlapped at all 11 early checkpoints, so the proposed detector could not work. The claims gate reports 6 supported, 2 unsupported and 1 underpowered claim—and keeps the failures visible.